

1.6 TB

The size of the largest SSD made by BitMICRO

100 GB

The data you can download in a day on a 10 Mbit line.

TRIED & TESTED

Internal desktop hard drives

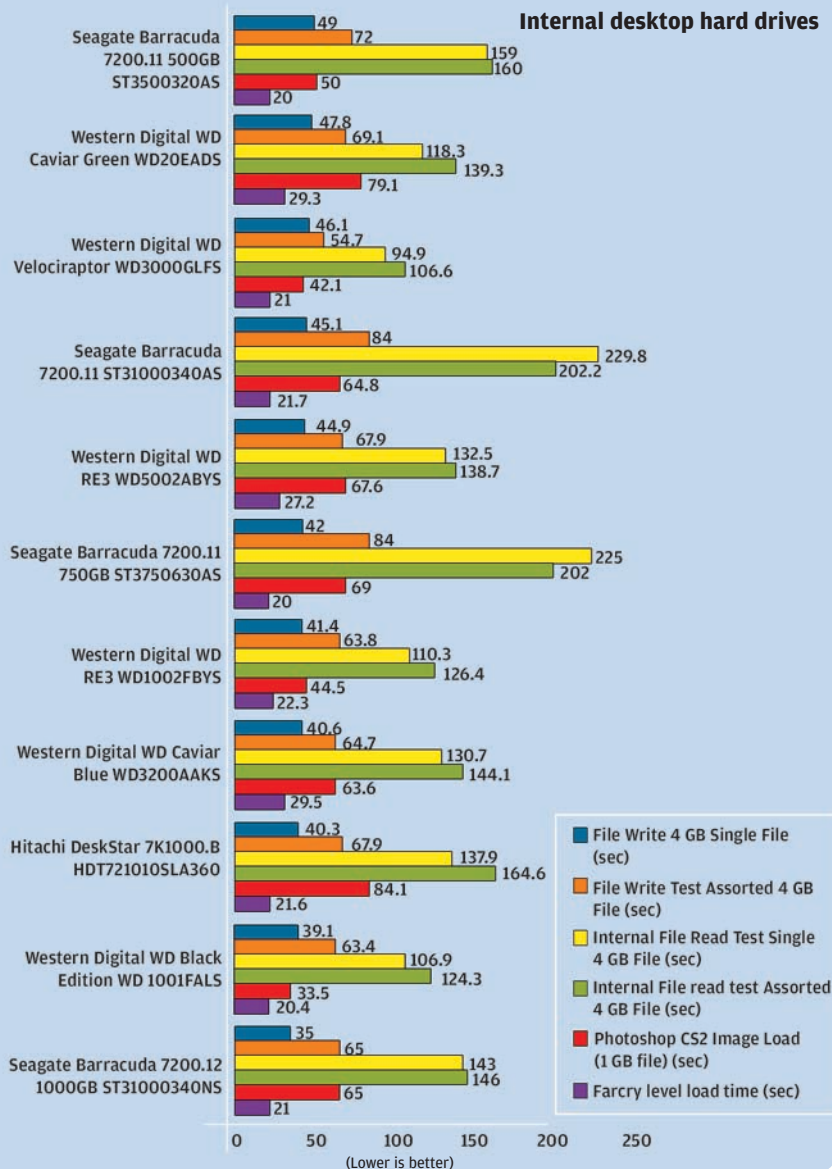
Internal hard drives are the driving force for the storage market. Hard drives are one of the cheapest ways of storing large amounts of data. With capacities soaring and prices plummeting, the cost of data storage keeps falling. This year, we received some 15 internal drives in this category with the largest being the WD Caviar Green 2 TB drive. The next largest capacity was that of the Seagate 7200.11 1.5 TB drive.

The expected winner in performance was the WD Velociraptor with a ridiculous overall score of 81.77. It's amazing to know that all this performance comes from a drive the size of a notebook hard drive with a heat-sink bolted atop for cooling, and that you can fix it in your computer case. Although not the fastest in sequential file writing speeds, the drive achieved most of its score through the assorted file write tests and the internal file transfer tests.



The other drives could only hope to compete with the Velociraptor in tests such as the *Far Cry* level loading run. The drive to load it the fastest was surprisingly the ones from the Seagate's 7200.11 series. The all new 7200.12 1 TB drive also matched Velociraptor's score at this test. It's also the same drive that scored a really high burst read speed of 482.6 Mbps.

Generally looking at the scores we got from the tests, it appears as if the Seagate



How we tested

All the products were tested on a high-end configuration. A Gigabyte P45-based solution was used for the motherboard and the processor used was an Intel Core 2 Extreme QX9650. Four gigabytes of OCZ DDR2 Dominator memory was used. The operating system used was Windows XP and it ran on an Intel 80 GB Solid State Drive. A Western Digital Velociraptor was used alongside to dump data to, while read tests were performed on the test drive.

The first run of tests was the synthetic tests. These tests consisted of HDtach with 8 MB and 32 MB zones. SiSoft Sandra's disk performance benchmarks were also used. Write and read tests along with the access response time tests were run to get statistics for the drives. The weights in the overall scorings for these benchmarks used were lowered.

It's seen in the past with many tests and it's no different for hard drives: Good synthetic test results don't always translate to great performance in every application. Higher importance was given to the real-world tests instead. These tests consisted of simple file copy operations. A set of four gigabyte file and a folder with four gigabytes of data consisting of smaller files were used.

Two equally sized NTFS partitions were created on the test drive and files were copied to the partition. The internal file copy test included copying the same set of files from one partition to another. Photoshop was setup to use the test hard drive's partition as a scratch disk to dump temporary files. A 1 GB file from the same partition was opened and times recorded. For internal drives, *Far Cry* was installed on the test disk and the first level's load times were recorded.

For external drives, a read test was used where files from the external disk were copied to the WD Velociraptor. For the sake of fairness, all the external drives were rated with scores recorded over the USB interface. All the drives were put through the real-world tests using all other connectivity options available, which included Firewire and eSATA. The scores are displayed in the table at the end of the test for your reference.

Network storage drives were tested by copying the same set of data files used for the other disk drives over an isolated network to the network drive.